is evidence of the uniformity of change throughout the cosmos, and thus implies the uniformity of elapsed time as well. Just as the uniformity of earth's geological structure allows a geologist in America, and one in Africa, and another in Asia to agree on earth's history and age, the uniformity of cosmic evolution throughout all of space allows a physicist in the Milky Way galaxy, and one in the Andromeda galaxy, and another in the Tadpole galaxy to all agree on the *universe*'s history and age. Concretely, the homogeneous evolution of the universe means that a clock here, a clock in the Andromeda galaxy, and a clock in the Tadpole galaxy will, on average, have been subject to nearly identical physical conditions and hence will have ticked off time in nearly the same way. The homogeneity of space thus provides a universal synchrony.

While I have so far left out important details (such as the expansion of space, covered in the next section) the discussion highlights the core of the issue: time stands at the crossroads of symmetry. If the universe had perfect temporal symmetry—if it were completely unchanging—it would be hard to define what time even means. On the other hand, if the universe did not have symmetry in space—if, for example, the background radiation were thoroughly haphazard, having wildly different temperatures in different regions—time in a cosmological sense would have little meaning. Clocks in different locations would tick off time at different rates, and so if you asked what things were like when the universe was 3 billion years old, the answer would depend on whose clock you were looking at to see that those 3 billion years had elapsed. *That* would be complicated. Fortunately, our universe does not have so much symmetry as to render time meaningless, but does have enough symmetry that we can avoid such complexities, allowing us to speak of its overall age and its overall evolution through time.

So, let's now turn our attention to that evolution and consider the history of the universe.

Stretching the Fabric

The history of the universe sounds like a big subject, but in broad-brush outline it is surprisingly simple and relies in large part on one essential fact: The universe is expanding. As this is *the* central element in the